Tangents and normals

1 $m = -\frac{1}{2}$

2

a $y = 5\pi x - 5\pi^2$

b $y = -3\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$

3

a $x = \frac{\pi}{12}$

b $f'\left(\frac{\pi}{12}\right) = -4$

c $\theta = 104^\circ$

4 $\theta = 83^\circ$

5

a

i $\frac{dy}{dx} = 2\cos 2x$

ii Undefined

iii $x = \frac{\pi}{4}$

b

i $\frac{dy}{dx} = \sqrt{2}\cos x$

ii -1

iii $y = -x + \frac{\pi}{4} + 1$

c

i $\frac{dy}{dx} = -\sqrt{2}\sin x$

ii 1

iii $y = x - \frac{\pi}{4} + 1$

6

a $\frac{dy}{dx} = \frac{x\cos x - \sin x}{x^2}$

b $\frac{\pi^2}{4}$

c $y = \frac{\pi^2 x}{4} - \frac{\pi^3}{8} + \frac{2}{\pi}$

Trigonometric graphs

7

a $\frac{dy}{dx} = \frac{1}{1 + \cos x}$

b 0

8 $x = \frac{\pi}{\ }, \frac{5\pi}{\ }, \frac{9\pi}{\ }, \frac{13\pi}{\ }$

9

a $\left(\frac{\pi}{3}, -4\right), \left(\frac{4\pi}{3}, 4\right)$

b $f''(x) = 4\sin\left(x + \frac{\pi}{6}\right)$

c $\left(\frac{\pi}{3}, -4\right)$ is a minimum, and $\left(\frac{4\pi}{3}, 4\right)$ is a maximum

10

a

i $y = 0, x = 0, \pi, 2\pi$

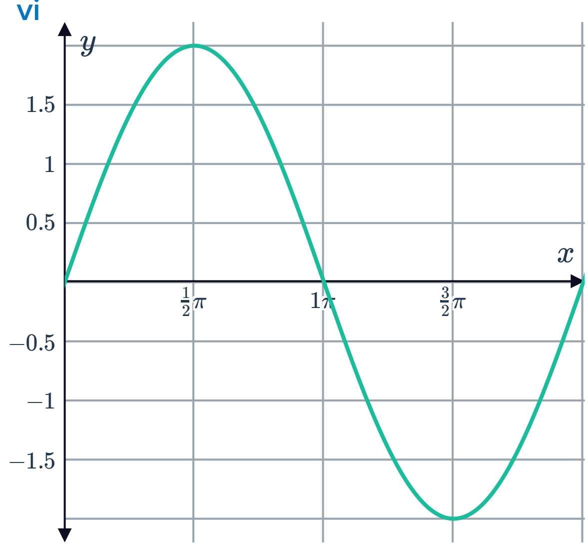
ii $f'(x) = 2\cos x$

iii $\left(\frac{\pi}{2}, 2\right), \left(\frac{3\pi}{2}, -2\right)$

iv $f''(x) = -2\sin x$

v Min at $\left(\frac{3\pi}{2}, -2\right)$, Max at $\left(\frac{\pi}{2}, 2\right)$

vi



b

i $y=1, x=\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

ii $f'(x)=-3\sin 3x$

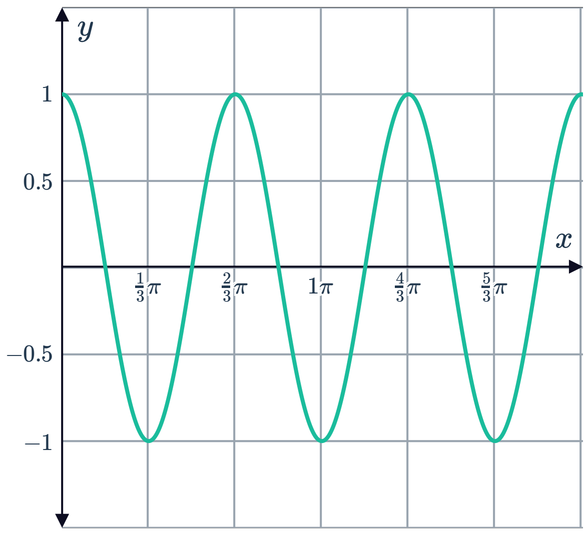
iii $\left(\frac{\pi}{3},-1\right), (\pi,-1), \left(\frac{5\pi}{3},-1\right), (0,1), \left(\frac{2\pi}{3},1\right), \left(\frac{4\pi}{3},1\right), (2\pi,1)$

iv $f''(x)=-9\cos 3x$

v Minimum at $\left(\frac{\pi}{3},-1\right), (\pi,-1), \left(\frac{5\pi}{3},-1\right)$

Maximum at $(0,1), \left(\frac{2\pi}{3},1\right), \left(\frac{4\pi}{3},1\right), (2\pi,1)$

vi



c

i $y=0, x=0, \frac{\pi}{2}, \frac{3\pi}{2}$

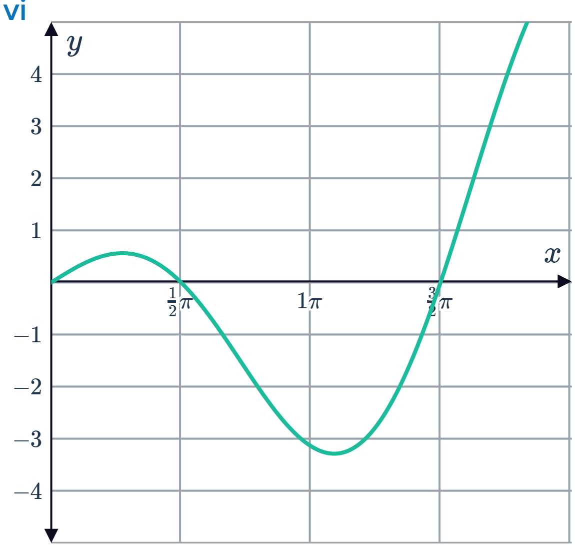
ii $f'(x)=\cos x - x\sin x$

iii $(0.86, 0.56), (3.42, -3.29)$

iv $f''(x)=-2\sin x - x\cos x$

v Minimum at $(3.42, -3.29)$
Maximum at $(0.86, 0.56)$

vi



d

i $y = 1, x = \frac{\pi}{2}, \frac{3\pi}{2}$

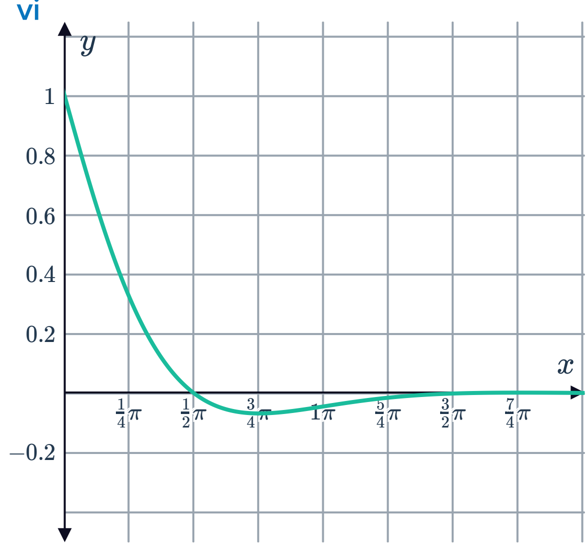
ii $f'(x) = -e^{-x}\sin x - e^{-x}\cos x$

iii $\left(\frac{3\pi}{4}, -\frac{e^{-\frac{3\pi}{4}}}{\sqrt{2}}\right)$

iv $f''(x) = 2e^{-x}\sin x$

v Minimum point

vi



e

i $y = 0, x = 0, \pi, 2\pi$

ii $f'(x) = 2\sin x\cos x$

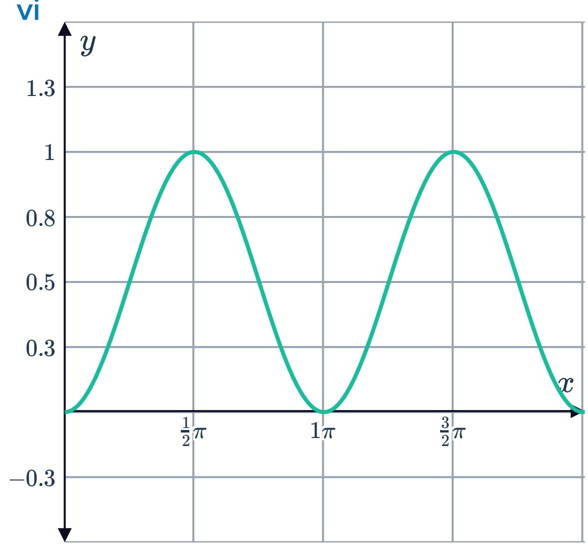
iii $(0,0), (\pi, 0), (2\pi, 0), \left(\frac{\pi}{2}, 1\right), \left(\frac{3\pi}{2}, 1\right)$

iv $f''(x) = 2\left(-\sin^2(x) + \cos^2(x)\right)$

v Minimum points: $(0,0), (\pi,0), (2\pi,0)$

Maximum points: $\left(\frac{\pi}{2}, 1\right), \left(\frac{3\pi}{2}, 1\right)$

vi



f

i $y=1, x=\frac{\pi}{2}, \frac{3\pi}{2}$

ii $f'(x)=-3\cos^2(x)\sin x$

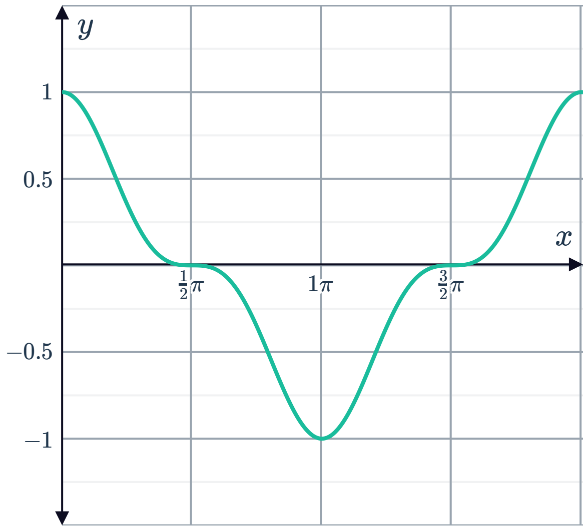
iii $(\pi,-1),(0,1),(2\pi,1),\left(\frac{\pi}{2},0\right),\left(\frac{3\pi}{2},0\right)$

iv $f''(x)=-3\cos x\left(\cos^2(x)-2\sin^2(x)\right)$

v Minimum point: $(\pi,-1)$, Maximum points: $(0,1),(2\pi,1)$

Horizontal points of inflection: $\left(\frac{\pi}{2},0\right),\left(\frac{3\pi}{2},0\right)$

vi



g

i $y=0, x=0$

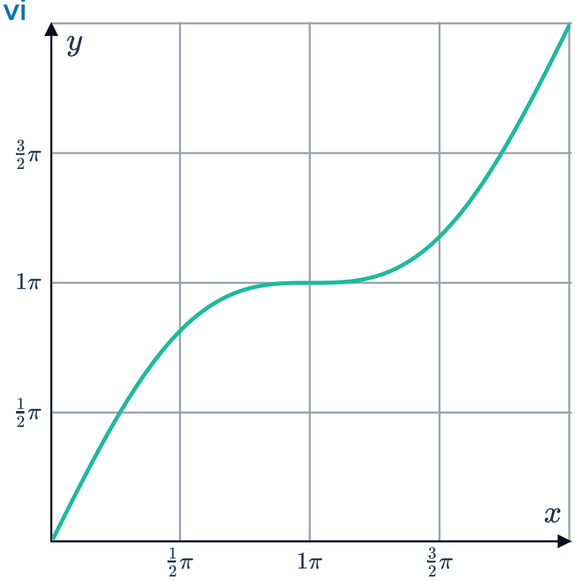
ii $f'(x)=1+\cos x$

iii (π,π)

iv $f''(x)=-\sin x$

v Horizontal point of inflection

vi



Applications

11 $k = \pm 5$

12

a It is 5 units to the left of the origin.

b $v\left(t\right) = 10 \sin 2t$

c No solution provided.

d -20 cm/s^2

13

a 0 cm

b $v\left(t\right) = -2\cos^2\left(t\right) + \cos t - 4$

c -5 cm/s

14

a 0 cm

b 16 cm

c $v\left(t\right) = 48\pi \cos 3\pi t$

d $t = \frac{1}{6}, \frac{1}{2}$

e 16 cm

f -16 cm

g 32 cm

15

a $v\left(t\right) = 76\pi \cos 4\pi t$

b $76\pi \text{ cm/s}$

c $t = \frac{1}{4}, \frac{1}{2}$

d The centre of its swing at $x = 0$.